

Mathematical Foundations for Computing

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Module 1

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Module 1: Sets and Set Theory

Module 2

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Module 2: Number Systems and Bases

Module 3

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Module 3: Functions and Their Properties

Module 4

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Module 4: Relations

Lesson 4.1 Understanding the concept of relations

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Definition of a relation: relation versus function

Video • 8 min

✔

Definition of a relation: relation versus function

Practice Assignment • Grade: 100%

✔

Matrix and graph representations of a relation

Video • 10 min

✔

Matrix and graph representations of a relation

Practice Assignment • Grade: 100%

✔

The properties of a relation: reflexive, symmetric and anti-symmetric

Video • 14 min

✔

The properties of a relation: reflexive, symmetric and anti-symmetric

Practice Assignment • Grade: 100%

✔

Relation properties: transitivity

Video • 5 min

✔

Relation properties: transitivity

Practice Assignment • Grade: 100%

Lesson 4.2 Equivalence, and partial and total order relations

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Equivalence relations and equivalence classes

Video • 6 min

✔

Equivalence relations and equivalence classes

Practice Assignment • Grade: 100%

✔

Partial and total order

Video • 6 min

✔

Partial and total order

Practice Assignment • Grade: 100%

✔

Building and analysing different types of mathematical relations

Dialogue • 15 min

Lesson 4.3 Summary and Assessments

✔

Topic summary

Reading • 15 min

✔

Check your understanding: End of module 4

Graded Assignment • Grade: 96%

Course Wrap-Up

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Mathematical Foundations for Computing: Course Summary

Reading • 10 min

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The Archive Room: A Mathematical Escape Mystery

Role Play • 15 min

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Course summary

Video • 1 min

Mathematical Foundations for Computing: Course Summary

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Congratulations on successfully completing the '**Mathematical Foundations for Computing**' course! We are incredibly proud of your dedication and hard work throughout this learning journey.

As we reach the end of our exploration of the essential mathematical concepts that underpin computer science, it's time to recap the key milestones, cement your understanding, and reflect on everything you've learned.

Sets and Set Theory: You began by mastering the fundamental concept of sets, learning to define sets and their elements, calculate cardinality, and perform operations such as union, intersection, and difference. You've developed skills in visualizing set relationships through Venn diagrams and applied De Morgan's laws using membership tables—techniques essential for understanding data structures and logical operations in computing.

Number Systems and Bases: You explored different number bases crucial to computing, particularly binary representation. You learned to convert integers and non-integers between decimal, binary, and other bases, and performed operations with binary numbers. This knowledge forms the foundation of understanding how computers represent and process data at the hardware level.

Functions and Their Properties: You investigated functions as mathematical models for computational processes, distinguishing between injective, surjective, and bijective functions. You analyzed special functions including logarithmic, floor, and ceiling functions, and mastered function composition—concepts directly applicable to algorithm design and analysis.

Relations: You concluded your journey by exploring relations and their properties, learning to represent them using matrices and digraphs. You examined reflexive, symmetric, anti-symmetric, and transitive relations, and applied concepts of equivalence relations and partial/total ordering to computational systems and data organization.

Throughout this course, you have not only gained a comprehensive theoretical understanding of mathematical foundations but also developed practical skills in applying these concepts to computing contexts. You've learned to see the connections between abstract mathematics and concrete computational problems, building a toolkit that will serve you throughout your computer science studies and career.

This course has equipped you with the essential mathematical language and reasoning skills needed to excel in more advanced topics within the Essential Mathematics for Computer Science specialisation. The formal structures you've mastered provide the foundation for logic, algorithms, and computational complexity that you'll encounter in future studies.

If you enjoyed this course, we invite you to leave a review or provide feedback. Your insights are invaluable and will help us enhance the learning experience for future students.

Once again, congratulations on reaching this milestone, and thank you for being part of this learning journey. The mathematical foundations you've built will transform your approach to computational problem-solving and algorithm design.

Best of luck as you continue your journey in computer science!

Go to next item

✔ Completed

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